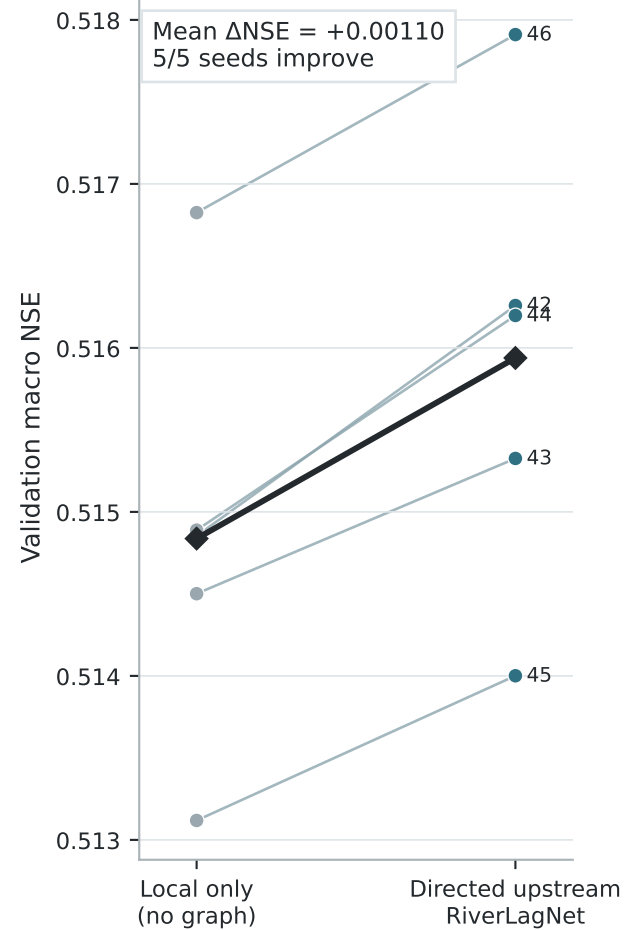


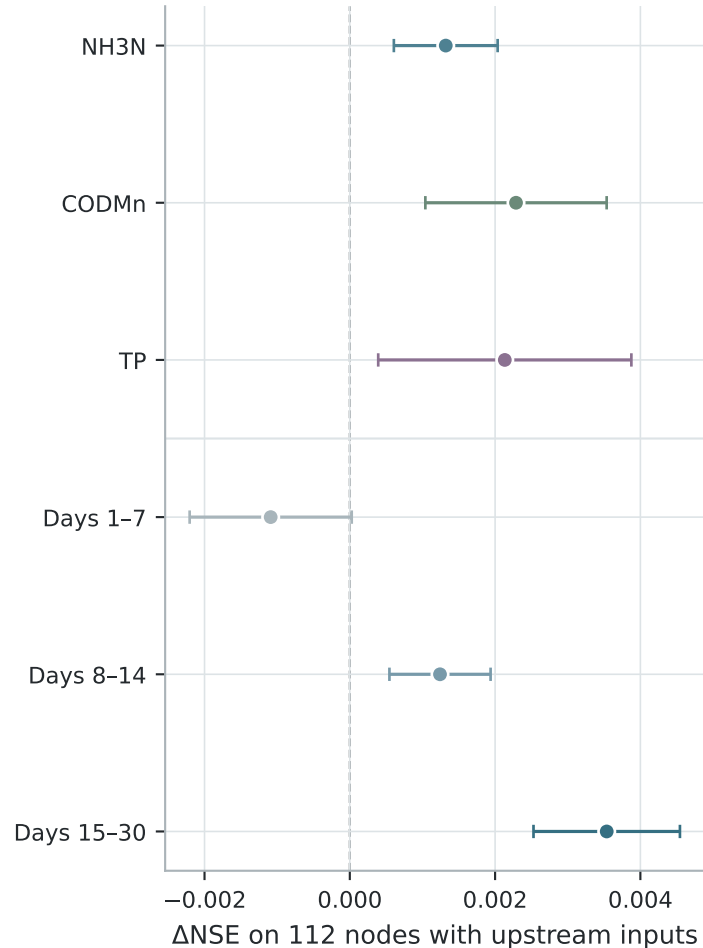
Directed upstream information yields a consistent validation gain

Five paired seeds · 238 monitored segments · 237 directed edges · local forecaster frozen before graph-residual training

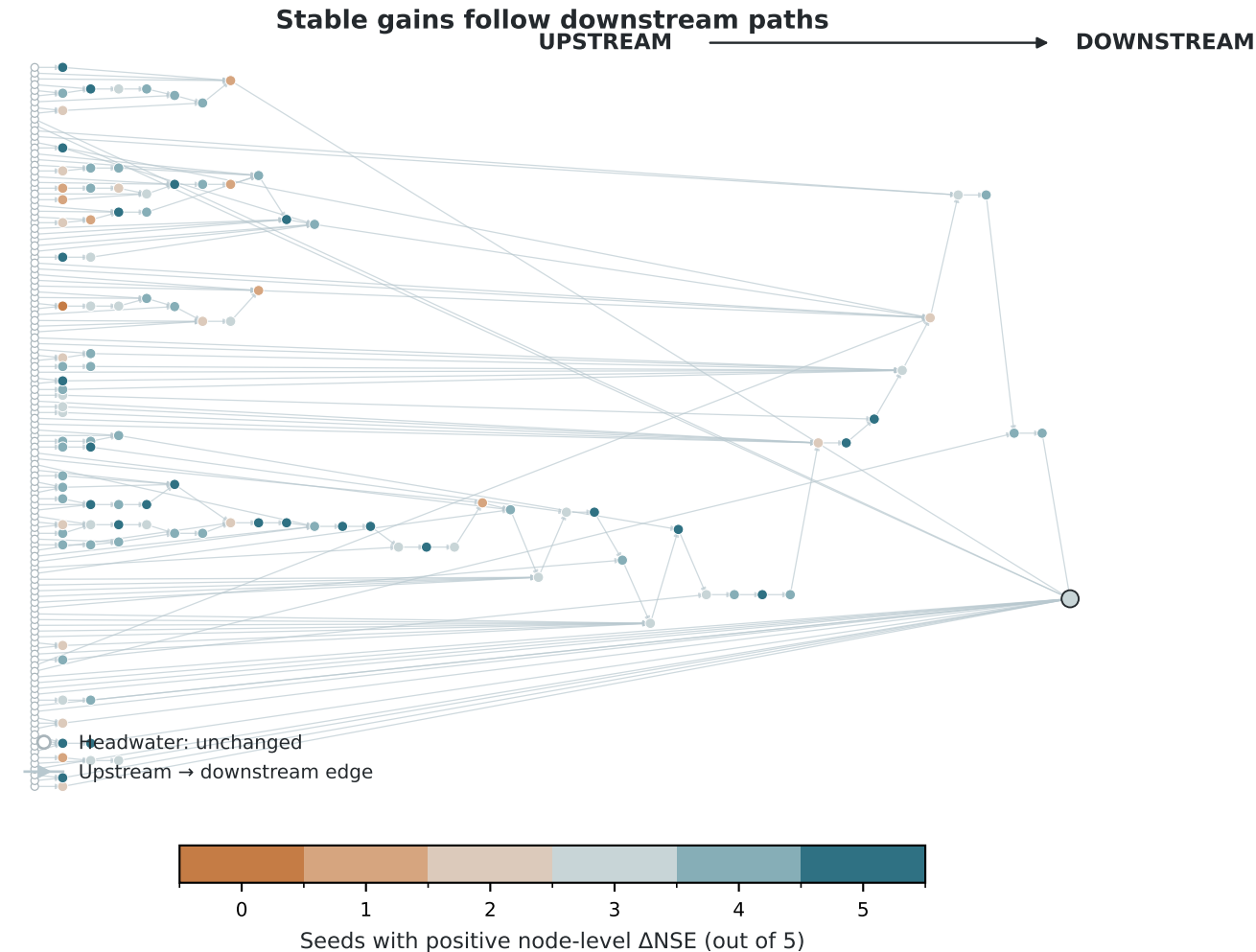
a Consistent across seeds



b Concentrated at longer leads



c



Validation split only; the held-out test set remains unopened. Node colors show predictive increments, not causal effects.